SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 3

ARMS PORTAL – NEGATIVE TEST SCENARIOS AND TEST CASES

Aim:

To write a case study to identify and analyze the requirements of ARMS Portals, and prepare negative test scenarios as well as test cases for the application.

1. Problem Description

The ARMS Portal is an academic/administrative web portal through which authorized students and staff can sign in, view academic information, submit requests, upload documents and track request status. The case study assumes these common portal functions; institution-specific rules should be confirmed from the official ARMS specification.

The ARMS Portal is expected to provide reliable, secure and user-friendly services. The testing team verifies the stated requirements using valid and invalid inputs, records actual results, and reports any deviation from the expected behaviour.

2. Functional Requirements

1. Students and staff must sign in using valid institutional credentials.
2. The portal must display information only to the authenticated user with the appropriate role.
3. Students must be able to view their profile and academic details.
4. Users must be able to submit permitted requests with required information.
5. The portal must validate uploaded documents for file type and size.
6. Submitted requests must receive a unique reference number.
7. Users must be able to view the status of their own requests.
8. The portal must display clear validation and access-denied messages.

4. Negative Test Scenarios and Test Cases

Negative testing checks whether invalid, incomplete, unauthorized or boundary inputs are safely rejected with a clear message.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Login with wrong password	Valid student ID with incorrect password	Login is denied with a clear error.	Pass
NT02	Submit empty request	Leave mandatory request fields blank	Required-field messages are displayed.	Pass
NT03	Access another student's profile	Change URL or request another student ID	Access is denied; private data is not disclosed.	Pass
NT04	Upload unsupported file	Upload an executable file	File is rejected with supported formats listed.	Pass
NT05	Upload oversized document	Upload a file above the configured limit	Upload is blocked with a size message.	Pass
NT06	Use expired session	Submit a request after session expiry	User is redirected to login and no request is submitted.	Pass
NT07	Use invalid reference number	Enter a nonexistent request ID	No unrelated request details are shown.	Pass
NT08	Unauthorized staff action	Student attempts staff-only approval function	Operation is denied.	Pass

5. Detailed Execution of a Failed Test Case

Test Case ID: NT05

Precondition: The application is running in the test environment and the relevant test data is available.

Test Steps: 1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Upload a file above the configured limit. 4. Observe the application response.

Expected Result: Upload is blocked with a size message.

Actual Result: For demonstration, assume the application accepts the invalid input or does not display the required validation message.

Conclusion: The test is marked as failed for defect-report demonstration. This is an illustrative result, not a claim about a deployed application.

6. Sample Bug Report

Field	Description
Bug ID	BR-003
Linked Test Case	NT05
Severity	High (illustrative)
Priority	High (illustrative)
Bug Title	ARMS Portal does not correctly validate the condition: Upload oversized document.
Environment	Test environment; browser and operating system should be recorded during actual execution.
Preconditions	The application is running in the test environment and the relevant test data is available.
Steps to Reproduce	1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Upload a file above the configured limit. 4. Observe the application response.
Expected Result	Upload is blocked with a size message.
Actual Result	For demonstration, assume the application accepts the invalid input or does not display the required validation message.
Impact	Invalid data or an unauthorized operation may be accepted, resulting in incorrect processing, data integrity issues or a poor user experience.
Suggested Fix	Apply validation at both client and server sides, return a clear error message, and add regression tests for the boundary and invalid condition.
Status	Open – illustrative sample.

7. Bug Life Cycle

New → Assigned → In Progress → Fixed → Retesting → Closed. If the defect is still reproducible after the correction, the tester reopens it and returns it to the development team.

Conclusion

The test cases prepared for the ARMS Portal provide coverage of its key functional requirements. Executing them helps identify functional gaps, improve error handling and increase confidence in the quality of the application.